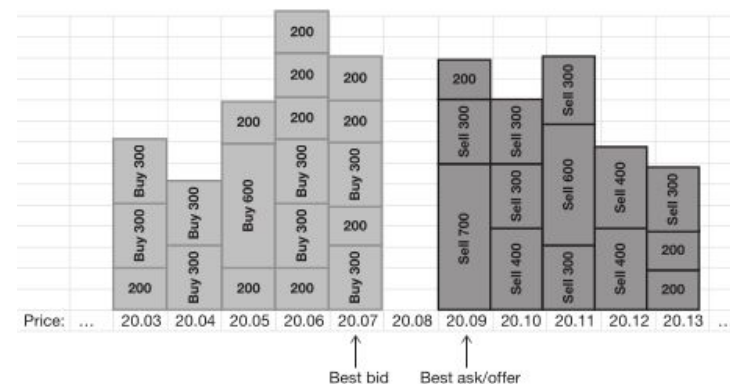


High-Frequency Trading: An Insider's Perspective

Quant The Rocket Scientist of Wall Street

High-Frequency Trading model building process

- Ideas
 - From any paper, your research or observe market behavior
- Select a model that is suitable for your idea and your resources
 - See assumptions and limitations of the model
- Select an asset class that is properly for your model
 - Look and determine whether the micro-level data, on average, matches the model assumptions or not
 - Mostly the properties of the micro-level in each asset class tend to behave similarly
- Model
 - Be Careful because most idea research from papers uses US, EU and JP market data
 - You have to adjust your method to suitable your market



Aldridge, Irene. *High-frequency trading: a practical guide to algorithmic strategies and trading systems*. Vol. 604. John Wiley & Sons, 2013. Figure 3.1; page 40.

High-Frequency Trading model building process

- Design execution
 - How to react to all market events and take action to reduce your net exposure to 0 as quickly as possible
- Design risk control
 - Such as position limits, wash sale, naked short, market impact, etc
- Backtesting
 - You should design your backtesting process like a realistic market
 - Check the log and, validate your execution reacts correctly to every scenario
- Improvement
 - What did you miss and not expect?
 - Limitation of model

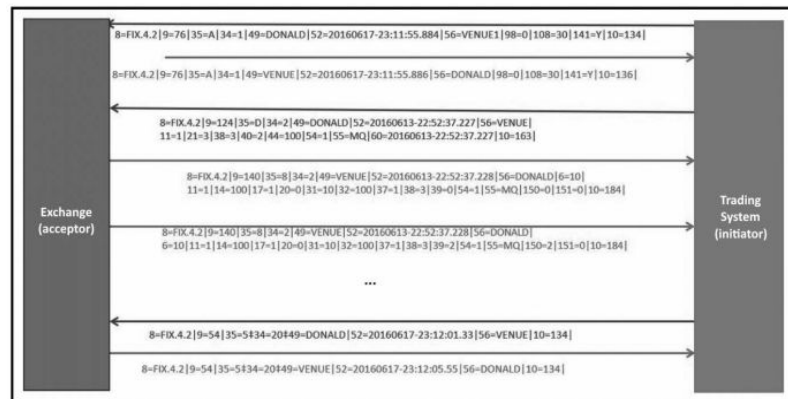


Figure 5.16 – A trading system using the FIX protocol to send an order to the exchange

Donadio, Sebastien, Sourav Ghosh, and Romain Rossier. *Developing High-Frequency Trading Systems: Learn how to implement high-frequency trading from scratch with C++ or Java basics*. Packt Publishing Ltd, 2022.

High-Frequency Trading model

- Market making

$$S^{b*} = S^{ref} - \delta^*/2$$

$$S^{a*} = S^{ref} + \delta^*/2$$

Where

S^{b*}, S^{a*} are the bid and ask fair prices

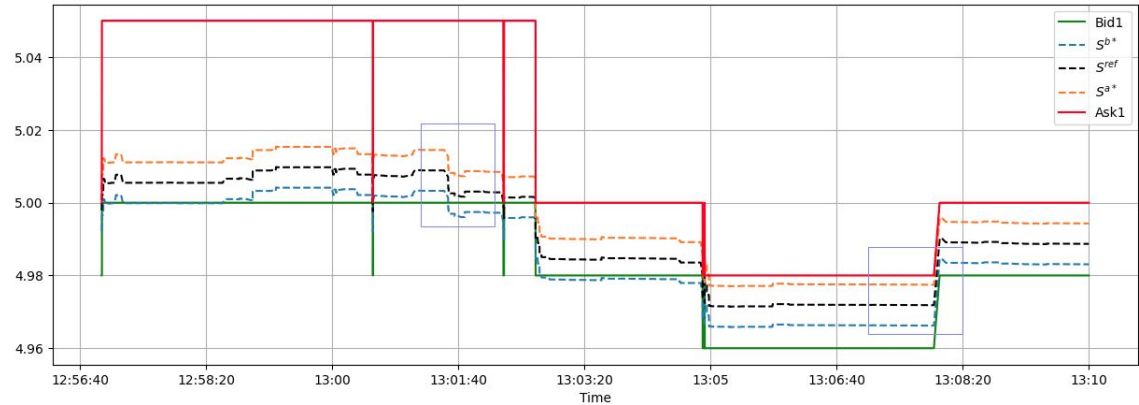
- Your risk such as inventory, balance, etc

S^{ref} is reference price

- Expected future price, short-term events

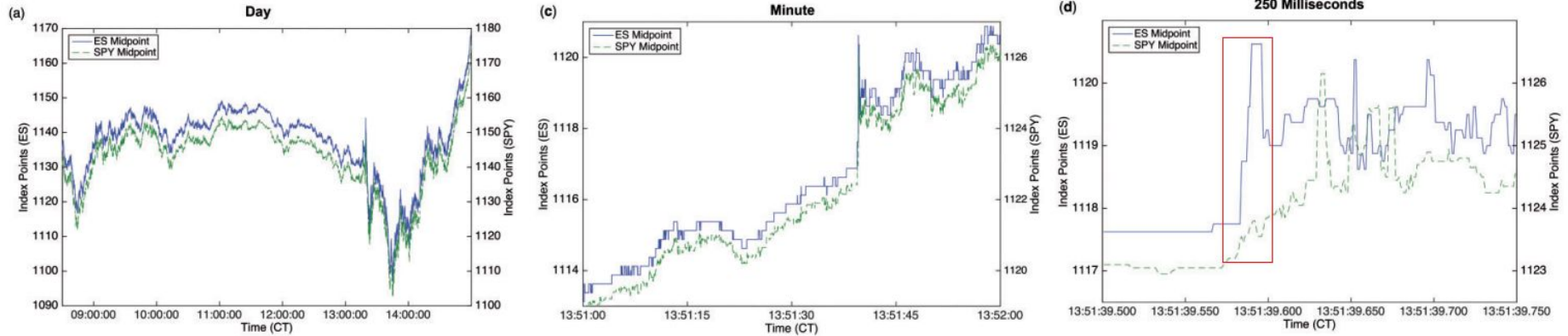
δ^* is expected spread

- Liquidity, volatility and adverse selection cost



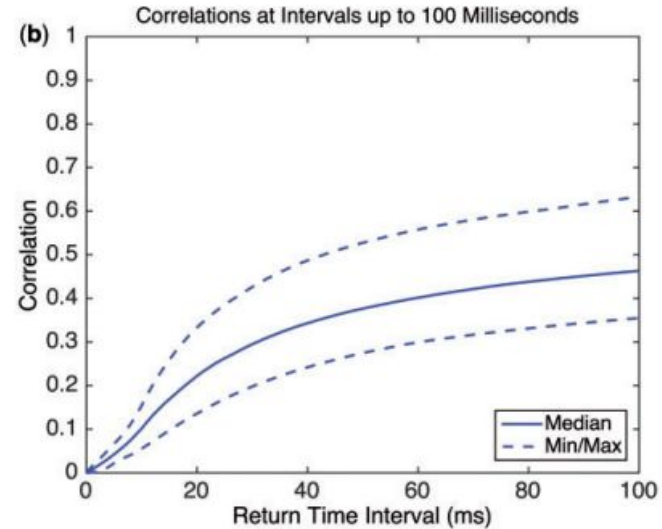
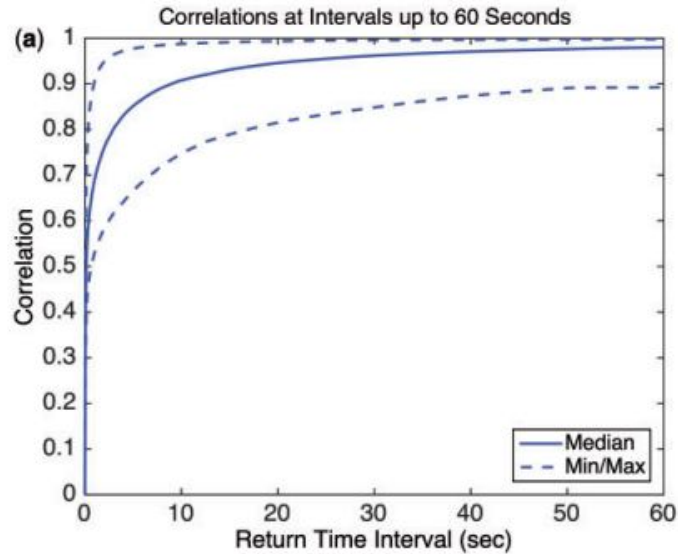
High-Frequency Trading model

- Arbitrage



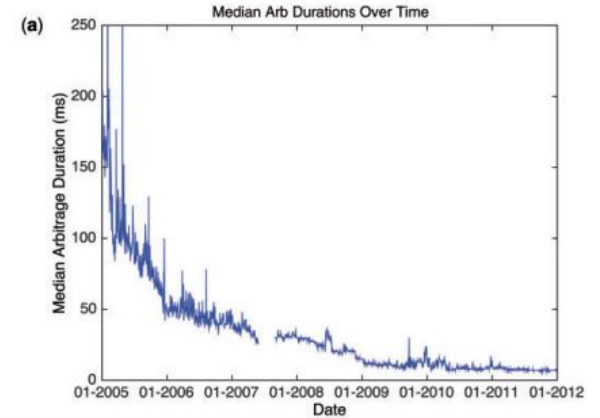
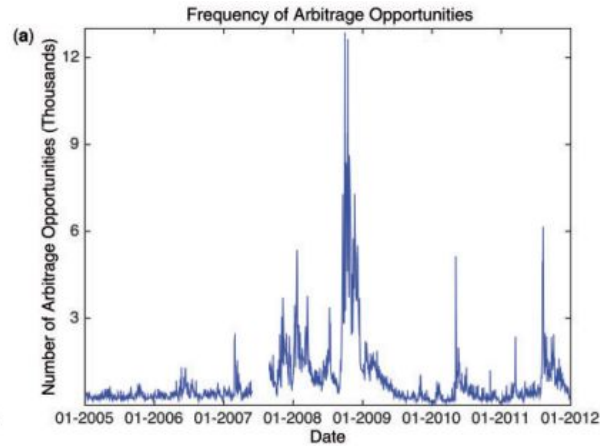
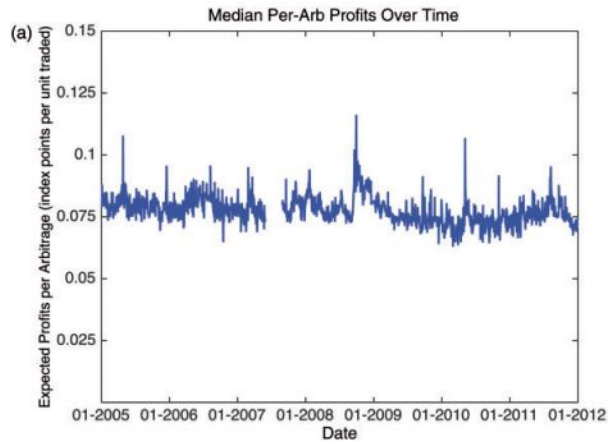
Budish, Eric, Peter Cramton, and John Shim. "The high-frequency trading arms race: Frequent batch auctions as a market design response." *The Quarterly Journal of Economics* 130.4 (2015): 1547-1621.

Correlation Breakdown



Budish, Eric, Peter Cramton, and John Shim. "The high-frequency trading arms race: Frequent batch auctions as a market design response." *The Quarterly Journal of Economics* 130.4 (2015): 1547-1621.

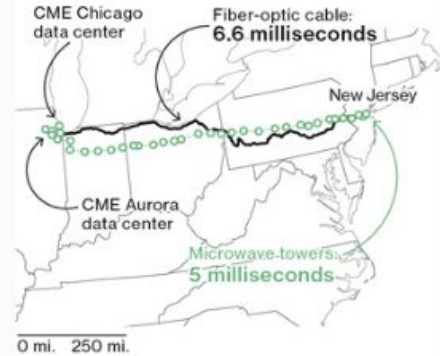
ES & SPY Arbitrage Opportunities over Time: 2005–2011



Budish, Eric, Peter Cramton, and John Shim. "The high-frequency trading arms race: Frequent batch auctions as a market design response." *The Quarterly Journal of Economics* 130.4 (2015): 1547-1621.

Light Speed Trading

A Jump affiliate owns a network of microwave towers that can convey data between New Jersey and Illinois about 1.6 milliseconds faster than fiber.

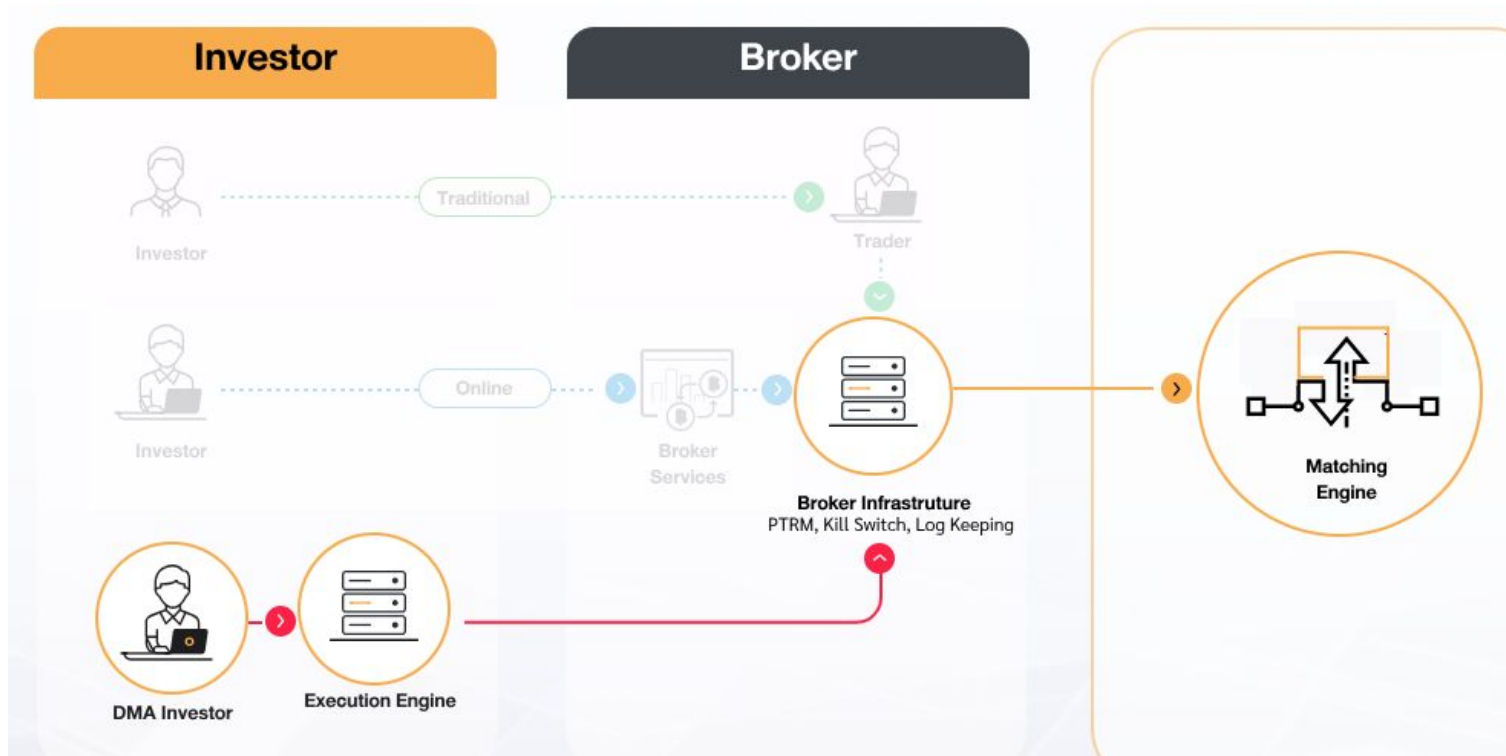


GRAPHIC BY BLOOMBERG BUSINESSWEEK. DATA: FCC, "INFORMATION TRANSMISSION BETWEEN FINANCIAL MARKETS IN CHICAGO AND NEW YORK" BY LAUGHLIN ET AL



<https://avelacom.com/connectivity-map>

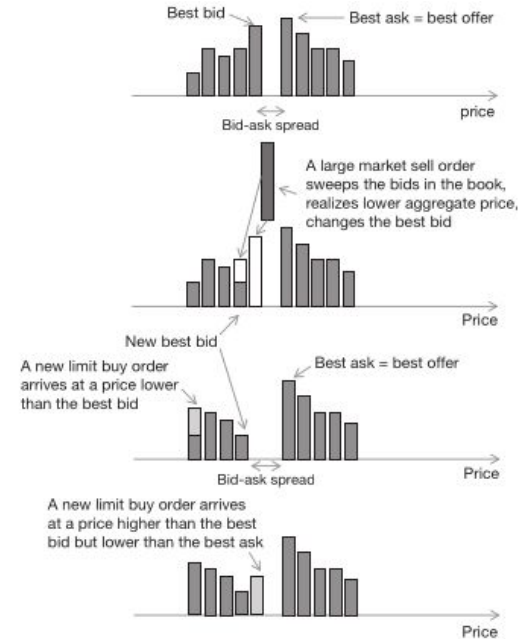
Dobrev, D., & Schaumburg, E. (2017). *High-Frequency Cross-Market Trading: Model Free Measurement and Applications*. Presented at the Alan Turing Institute Workshop on "Algorithmic Trading: Perspectives from Mathematical Modelling," March 1, 2017.



<https://www.set.or.th/th/services/institutions/direct-market-access>

What do you need to learn if you want to do HFT?

- Electronic exchange
 - How does the matching engine algorithm work? (time-priority, pro-rata)
 - Type of orders (limit, market, time in force, etc...)



Aldridge, Irene. *High-frequency trading: a practical guide to algorithmic strategies and trading systems*. Vol. 604. John Wiley & Sons, 2013. Figure 3.2; page 40.

What do you need to learn if you want to do HFT?

- Electronic exchange
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 - Type of orders (limit, market, time in force, etc...)
 - How does electronic exchange communicate?



Figure 5.15 – Trading system sending an order to the exchange

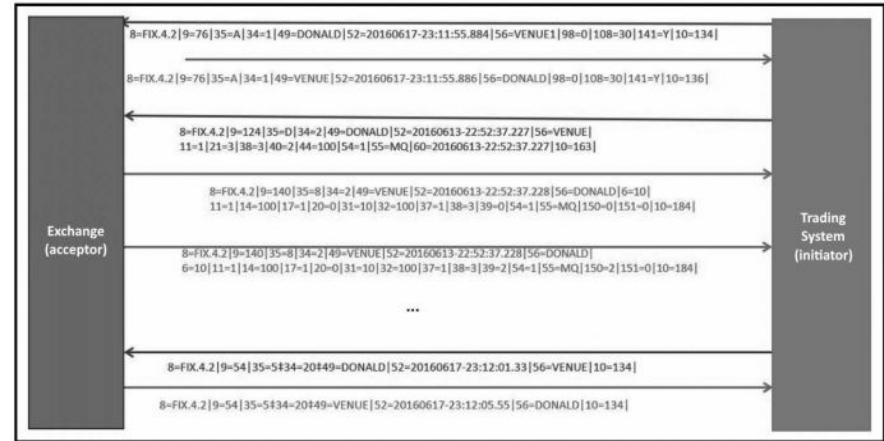


Figure 5.16 – A trading system using the FIX protocol to send an order to the exchange

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What do you need to learn if you want to do HFT?

- Electronic exchange
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 - Type of orders (limit, market, ice berg, time in force, etc...)
 - How electronic exchange communicate?
- Quant
 - Statistics and probability theory
 - Linear algebra
 - Market microstructure

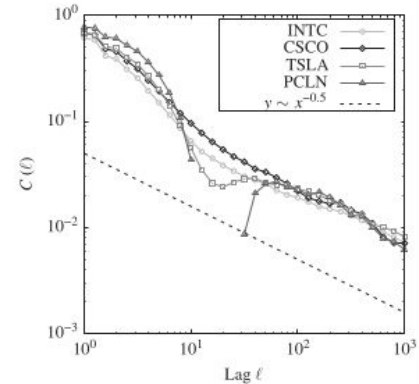


Figure 10.1. Autocorrelation function of the market order sign process for INTC, CSCO, TSLA and PCLN. The dashed line represents a power-law with exponent -0.5 . The missing data points correspond to negative values.

Bouchaud, Jean-Philippe, et al. *Trades, quotes and prices: financial markets under the microscope*. Cambridge University Press, 2018.

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 - Market microstructure
 - Optimization
 - Partial differential equation
 - Stochastic optimal control (seems so scary but if you get a result like the images on the right-hand side, it's pretty cool, right? lol)
 - High school math (but you need speed! so you need a very good infrastructure)

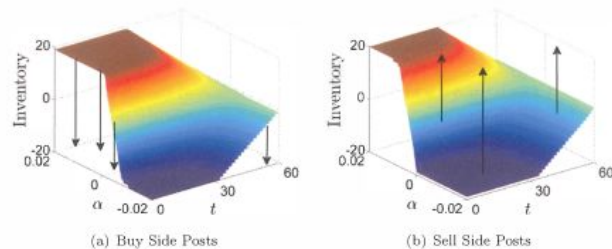
Thus the MM's value function satisfies the DPE

$$\begin{aligned} 0 = & \partial_t H + \frac{1}{2} \sigma^2 \partial_{SS} H - \phi q^2 \\ & + \lambda^+ \sup_{\delta^+} \left\{ e^{-\kappa^+ \delta^+} \mathbb{1}_{q > \bar{q}} \mathbb{E} [H(t, x + (S + \delta^+), S + \epsilon^+, q - 1) - H] \right. \\ & \quad \left. + (1 - e^{-\kappa^+ \delta^+}) \mathbb{1}_{q > \bar{q}} \mathbb{E} [H(t, x, S + \epsilon^+, q) - H] \right\} \\ & + \lambda^- \sup_{\delta^-} \left\{ e^{-\kappa^- \delta^-} \mathbb{1}_{q < \bar{q}} \mathbb{E} [H(t, x - (S - \delta^-), S - \epsilon^-, q + 1) - H] \right. \\ & \quad \left. + e^{-\kappa^- \delta^-} \mathbb{1}_{q < \bar{q}} \mathbb{E} [H(t, x, S - \epsilon^-, q) - H] \right\}, \end{aligned} \quad (10.23)$$

subject to the terminal condition

$$H(T, x, S, q) = x + q(S - \alpha q), \quad (10.24)$$

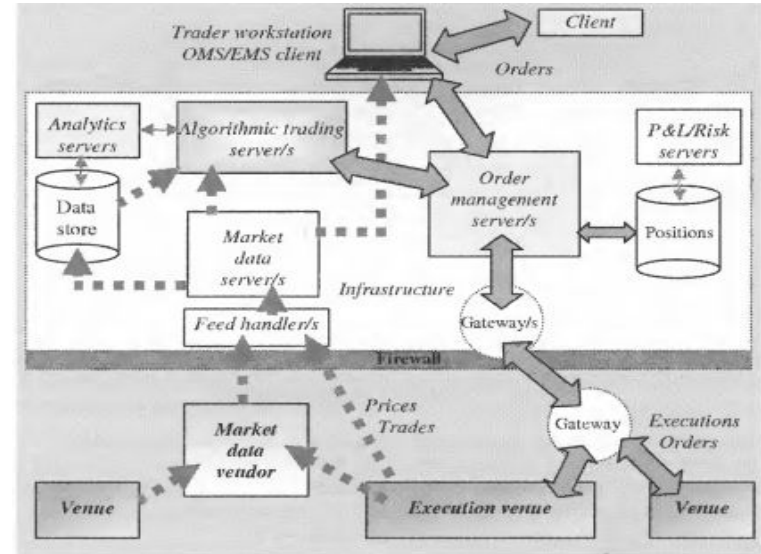
and where the expectation is over the random variables $\epsilon^\pm$ (not over x , S , or q) and $\mathbb{1}$ is the indicator function.



Cartea, Álvaro, Sebastian Jaimungal, and José Penalva. *Algorithmic and high-frequency trading*. Cambridge University Press, 2015. Chapter 10 Market making.

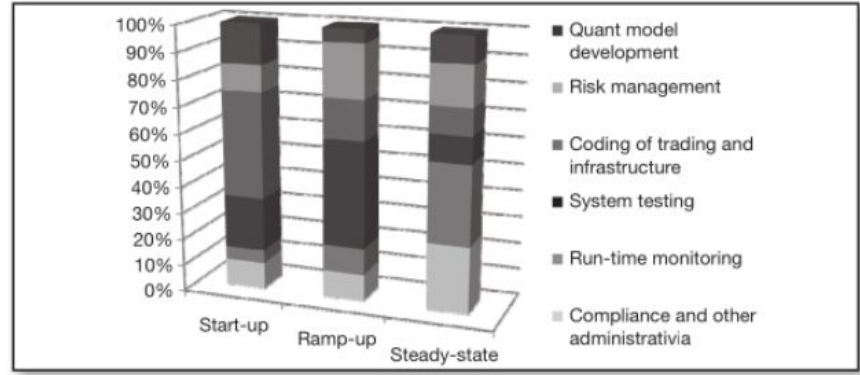
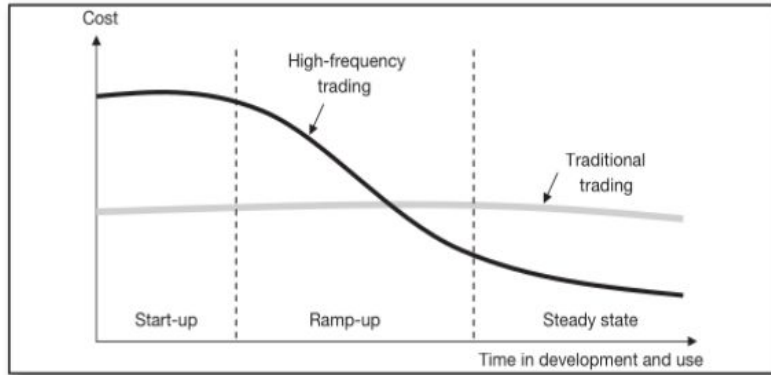
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 - Stochastic optimal control
 - High school math
- Engineering
 - FIX, ITCH/OUCH protocol
 - Computer networking
 - Data structure & algorithms
 - High performance computing
 - Operating system
 - Computer architecture
 - FPGA



Johnson, Barry. "Algorithmic Trading & DMA: An introduction to direct access trading strategies." (No Title) (2010). Figure 11-6; page 323.

The Business of High-Frequency Trading



Aldridge, Irene. *High-frequency trading: a practical guide to algorithmic strategies and trading systems*. Vol. 604. John Wiley & Sons, 2013. Figure 3.2; page 40.